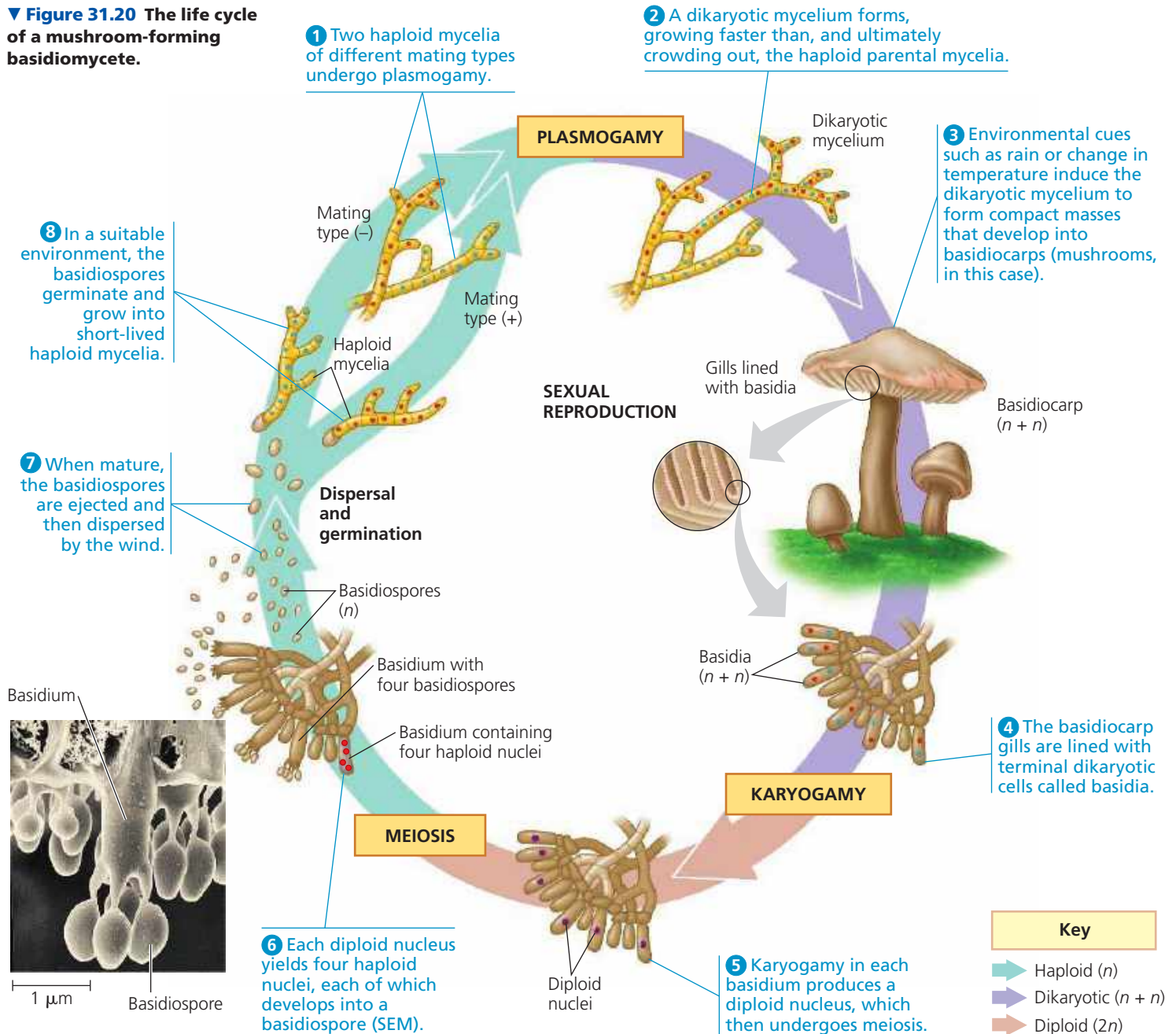


▼ **Figure 31.20** The life cycle of a mushroom-forming basidiomycete.



VISUAL SKILLS Use the diagram to determine the ploidy of a cell in the aboveground stalk of a mushroom.

➔ **Mastering Biology Animation: Life Cycle of a Mushroom**

elaborate fruiting bodies called **basidiocarps** (Figure 31.20). The common white mushrooms in the supermarket are familiar examples of a basidiocarp.

By concentrating growth in the hyphae of mushrooms, a basidiomycete mycelium can erect its fruiting structures in just a few hours; a mushroom pops up as it absorbs water and as cytoplasm streams in from the dikaryotic mycelium. By this process, in some species a ring of mushrooms, popularly called a “fairy ring,” may appear literally overnight (Figure 31.21). The mycelium below the fairy ring expands outward at a rate of about 30 cm per year, decomposing

organic matter in the soil as it grows. Some giant fairy rings are produced by mycelia that are centuries old.

After a mushroom forms, its cap supports and protects a large surface area of dikaryotic basidia on gills. During karyogamy, the two nuclei in each basidium fuse, producing a diploid nucleus (see Figure 31.20). This nucleus then undergoes meiosis, yielding four haploid nuclei, each of which ultimately develops into a basidiospore. Large numbers of basidiospores are produced: The gills of a common white mushroom have a surface area of about 200 cm² and may drop a billion basidiospores, which blow away.